RepPoints: Point Set Representation for Object Detection

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Abstract

*Modern object detectors rely heavily on rectangular bounding boxes, such as anchors, proposals and the final predictions, to represent objects at various recognition stages. The bounding box is convenient to use but provides only a coarse localization of objects and leads to a correspondingly coarse extraction of object features. In this paper, we present **RepPoints** (representative points), a new finer representation of objects as a set of sample points useful for both localization and recognition. Given ground truth localization and recognition targets for training, RepPoints learn to automatically arrange themselves in a manner that bounds the spatial extent of an object and indicates semantically significant local areas. They furthermore do not require the use of anchors to sample a space of bounding boxes. We show that an anchor-free object detector based on RepPoints can be as effective as the state-of-the-art anchor-based detection methods, with 46.5 AP and 67.4 AP_{50} on the COCO test-dev detection benchmark, using ResNet-101 model. Code is available at https://github.com/microsoft/RepPoints.*

1. Introduction

Object detection aims to localize objects in an image and provide their class labels. As one of the most fundamental tasks in computer vision, it serves as a key component for many vision applications, including instance segmentation [30], human pose analysis [37], and visual reasoning [41]. The significance of the object detection problem together with the rapid development of deep neural networks has led to substantial progress in recent years [7, 11, 10, 33, 14, 3].

In the object detection pipeline, bounding boxes, which encompass rectangular areas of an image, serve as the basic element for processing. They describe target locations

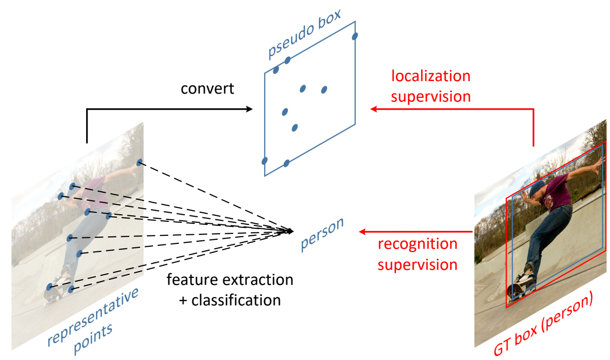

Figure 1. *RepPoints* is a new representation for object detection that consists of a set of points which indicate the spatial extent of an object and semantically significant local areas. This representation is learned via weak localization supervision from rectangular ground-truth boxes and implicit recognition feedback. Based on the richer RepPoints representation, we develop an anchor-free object detector that yields improved performance compared to using bounding boxes.

of objects throughout the stages of an object detector, from anchors and proposals to final predictions. Based on these bounding boxes, features are extracted and used for purposes such as object classification and location refinement. The prevalence of the bounding box representation can partly be attributed to common metrics for object detection performance, which account for the overlap between estimated and ground truth bounding boxes of objects. Another reason lies in its convenience for feature extraction in deep networks, because of its regular shape and the ease of subdividing a rectangular window into a matrix of pooled cells.

Though bounding boxes facilitate computation, they provide only a coarse localization of objects that does not conform to an object's shape and pose. Features extracted from the regular cells of a bounding box may thus be heavily influenced by background content or uninformative foreground areas that contain little semantic information. This may result in lower feature quality that degrades classification performance in object detection.

In this paper, we propose a new representation, called

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RepPoints：用于目标检测的点集表示方法

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摘要

Modern object detectors rely heavily on rectangular bounding boxes, such as anchors, proposals and the final predictions, to represent objects at various recognition stages. The bounding box is convenient to use but provides only a coarse localization of objects and leads to a correspondingly coarse extraction of object features. In this paper, we present ***RepPoints*** *(representative points), a new finer representation of objects as a set of sample points useful for both localization and recognition. Given ground truth localization and recognition targets for training, RepPoints learn to automatically arrange themselves in a manner that bounds the spatial extent of an object and indicates semantically significant local areas. They furthermore do not require the use of anchors to sample a space of bounding boxes. We show that an anchor-free object detector based on RepPoints can be as effective as the state-of-the-art anchor-based detection methods, with 46.5 AP and 67.4 AP_{50} on the COCO test-dev detection benchmark, using ResNet-101 model. Code is available at https://github.com/microsoft/RepPoints.*

1. 引言

目标检测旨在定位图像中的物体并提供其类别标签。作为计算机视觉中最基础的任务之一，它是许多视觉应用的关键组成部分，包括实例分割[30]、人体姿态分析[37]和视觉推理[41]。目标检测问题的重要性与深度神经网络的快速发展相结合，近年来取得了显著进展[7, 11, 10, 33, 14, 3]。

在目标检测流程中，包围盒作为处理的基本单元，它们框定图像中的矩形区域，用以描述目标位置。

*Equal contribution. †The work is done when Ze Yang and Shaohui Liu are interns at Microsoft Research Asia.

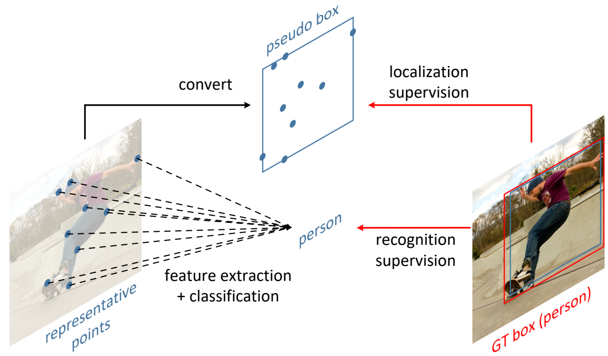

图1. *RepPoints* 是一种用于目标检测的新表征方式，它由一组指示物体空间范围及语义显著局部区域的点构成。该表征通过矩形标注框提供的弱定位监督与隐式识别反馈进行学习。基于这种更丰富的RepPoints表征，我们开发了一种无需锚框的目标检测器，相比使用边界框的方法取得了更优的性能。

在物体检测器的各个阶段，从锚框和候选框到最终预测，边界框始终贯穿其中。基于这些边界框，特征被提取并用于物体分类和位置细化等目的。边界框表示法的盛行，部分归因于物体检测性能的常用评估指标，这些指标考虑了物体估计边界框与真实边界框之间的重叠程度。另一个原因在于其便于深度网络中的特征提取，因为其规则形状以及将矩形窗口轻松细分为池化单元矩阵的特性。

尽管边界框便于计算，但它们仅能提供物体的粗略定位，无法贴合物体的形状和姿态。因此，从边界框的规则单元格中提取的特征可能会受到背景内容或缺乏语义信息的无意义前景区域的严重影响。这可能导致特征质量下降，从而降低物体检测中的分类性能。

在本文中，我们提出了一种新的表示方法，称为

RepPoints, that provides more fine-grained localization and facilitates classification. Illustrated in Fig. 1, RepPoints is a set of points that learns to adaptively position themselves over an object in a manner that circumscribes the object's spatial extent and indicates semantically significant local areas. The training of RepPoints is driven jointly by object localization and recognition targets, such that the RepPoints are tightly bound by the ground-truth bounding box and guide the detector toward correct object classification. This adaptive and differentiable representation can be coherently used across the different stages of a modern object detector, and does not require the use of anchors to sample over a space of bounding boxes.

RepPoints differs from existing non-rectangular representations for object detection, which are all built in a bottom-up manner [38, 21, 48]. These bottom-up representations identify individual points (e.g., bounding box corners or object extremities) and rely on handcrafted clustering to group them into object models. Their representations furthermore either are still axis-aligned like bounding boxes [38, 21] or require ground truth object masks as additional supervision [48]. In contrast, RepPoints are learned in a top-down fashion from the input image / object features, allowing for end-to-end training and producing fine-grained localization without additional supervision.

With RepPoints replacing all the conventional bounding box representations in a two-stage object detector, including anchors, proposals and final localization targets, we develop a clean and effective *anchor-free* object detector, which achieves 42.8 AP and 65.0 AP_{50} on the COCO benchmark [26] without multi-scale training and testing, and achieves 46.5 AP and 67.4 AP_{50} with multi-scale training and testing using ResNet-101 model. The proposed object detector not only surpasses all existing anchor-free detectors but also performing on-par with state-of-the-art anchor-based baselines.

2. Related Work

Bounding boxes for the object detection problem. The bounding box has long been the dominant form of object representation in the field of object detection. One reason for its prevalence is that a bounding box is convenient to annotate with little ambiguity, while providing sufficiently accurate localization for the subsequent recognition process. This may explain why the major benchmarks all utilize annotations and evaluations based on bounding boxes [8, 26, 20]. In turn, these benchmarks motivate object detection methods to use the bounding box as their basic representation in order to align with the evaluation protocols.

Another reason for the dominance of bounding boxes is that almost all image feature extractors, both before [39, 5] and during the deep learning era [19, 34, 36, 15], are based on an input patch with a regular grid form. It is thus convenient to use the bounding box representation to facilitate feature extraction [11, 10, 33].

Although the proposed *RepPoints* has an irregular form, we show that it can be amenable to convenient feature extraction. Our system utilizes RepPoints in conjunction with deformable convolution [4], which naturally aligns with RepPoints in that it aggregates information from input features at several sample points. Besides, a rectangular *pseudo box* can be readily generated from RepPoints (see Section 3.2), allowing the new representation to be used with object detection benchmarks.

Bounding boxes in modern object detectors. The best-performing object detectors to date generally follow a multi-stage recognition paradigm [24, 4, 13, 23, 2, 35, 27], and the bounding box representation appears in almost all stages: 1) as pre-defined [33, 25] or learnt [40, 45, 47] anchors that serve as hypotheses over the bounding box space; 2) as refined object proposals connecting successive recognition stages [33, 2]; and 3) as the final localization targets.

The RepPoints can be used to replace the bounding box representations in all stages of model object detectors, resulting in a more effective new object detector. Specifically, the anchors are replaced by center points, which is a special configuration of RepPoints. The bounding box proposals and final localization targets are replaced by the RepPoints proposals and final targets. Note the resulting object detector is anchor-free, due to the use of center points for initial object representation. It is thus even more convenient in use than the bounding box based methods.

Other representations for object detection. To address limitations of rectangular bounding boxes, there have been some attempts to develop more flexible object representations. These include an elliptic representation for pedestrian detection [22] and a rotated bounding box to better handle rotational variations [16, 49].

Other works aim to represent an object in a bottom-up manner. Early bottom-up representations include DPM [9] and Poselet [1]. Recently, bottom-up approaches to object detection have been explored with deep networks [21, 48]. CornerNet [21] first predicts top-left and bottom-right corners and then employs a specialized grouping method [28] to obtain the bounding boxes of objects. However, the two opposing corner points still essentially model a rectangular bounding box. ExtremeNet [48] is proposed to locate the extreme points of objects in the x- and y-directions [29] with supervision from ground-truth mask annotations. In general, bottom-up detectors benefit from a smaller hypothesis space (for example, CornerNet and ExtremeNet both detect 2-d points instead of directly detecting a 4-d bounding box) and potentially finer-grained localization. However, they have limitations such as relying on handcrafted clustering or post-processing steps to compose whole ob-

*RepPoints*它提供了更精细的定位并有助于分类。如图1所示，RepPoints是一组点，它们学习以自适应方式在物体上定位自身，从而勾勒出物体的空间范围并指示语义上重要的局部区域。RepPoints的训练由目标定位和识别任务共同驱动，使得RepPoints被紧密约束在真实标注边界框内，并引导检测器实现正确的物体分类。这种自适应且可微的表示能够连贯地应用于现代物体检测器的不同阶段，且无需使用锚框来对边界框空间进行采样。

RepPoints与现有的非矩形物体检测表示方法不同，后者均采用自下而上的方式构建[38, 21, 48]。这些自下而上的表示方法先识别独立点（如边界框角点或物体极值点），再依赖手工设计的聚类将其组合成物体模型。此外，它们的表示要么仍像边界框一样与坐标轴对齐[38, 21]，要么需要真实物体掩码作为额外监督[48]。相比之下，RepPoints通过自上而下的方式从输入图像/物体特征中学习，支持端到端训练，并能在无需额外监督的情况下实现精细定位。

通过将两阶段目标检测器中所有传统的边界框表示（包括锚点、提议和最终定位目标）替换为RepPoints，我们开发了一个简洁高效的*anchor-free*目标检测器。在COCO基准测试[26]中，该检测器无需多尺度训练和测试即可达到42.8 AP和65.0 AP_{50}；使用ResNet-101模型进行多尺度训练和测试时，性能进一步提升至46.5 AP和67.4 AP_{50}。所提出的目标检测器不仅超越了所有现有的无锚检测器，其性能也与当前最先进的基于锚点的基线模型相当。

2. 相关工作

目标检测问题中的边界框。边界框长期以来一直是目标检测领域中占主导地位的目标表示形式。其普遍存在的一个原因是，边界框标注方便且歧义性小，同时能为后续识别过程提供足够精确的定位。这或许解释了为什么主要基准测试都采用基于边界框的标注和评估方式[8, 26, 20]。反过来，这些基准测试也促使目标检测方法将边界框作为其基本表示形式，以便与评估标准保持一致。

边界框占据主导地位的另一个原因是，几乎所有图像特征提取器——无论是深度学习时代之前[39, 5]还是期间[19, 34, 36, 15]——都基于具有规则网格形式的输入图像块。因此它天使用边界框表示法便于特征提取[11, 10, 33]。

尽管提出的*RepPoints*具有不规则形态，我们证明其仍能适配便捷的特征提取。我们的系统将RepPoints与可变形卷积[4]结合使用，这种组合天然契合RepPoints的特性——后者通过多个采样点聚合输入特征信息。此外，矩形框*pseudo box*可直接从RepPoints生成（参见第3.2节），这使得新表征能够兼容现有目标检测基准框架。

现代物体检测器中的边界框。迄今为止性能最佳的物体检测器通常遵循多阶段识别范式[24, 4, 13, 23, 2, 35, 27]，而边界框表示几乎出现在所有阶段：1) 作为预定义[33, 25]或学习得到[40, 45, 47]的锚框，在边界框空间中充当假设；2) 作为连接连续识别阶段的优化物体提议[33, 2]；3) 作为最终的定位目标。

RepPoints可用于替换模型目标检测器所有阶段中的边界框表示，从而形成一个更有效的新目标检测器。具体来说，锚点被中心点所取代，这是RepPoints的一种特殊配置。边界框提议和最终定位目标被替换为RepPoints提议和最终目标。请注意，由于使用中心点进行初始目标表示，所得目标检测器是无锚点的。因此，它比基于边界框的方法使用起来更加方便。

目标检测的其他表示方法。为了解决矩形边界框的局限性，已有一些尝试开发更灵活的目标表示方法。这些方法包括用于行人检测的椭圆表示法[22]和能更好处理旋转变化的旋转边界框[16, 49]。

其他研究旨在以自下而上的方式表示物体。早期的自下而上表示方法包括DPM [9]和Poselet [1]。近年来，基于深度网络的自下而上物体检测方法得到了探索[21, 48]。CornerNet [21]首先预测左上角和右下角，然后采用专门的分组方法[28]来获取物体的边界框。然而，这两个相对的角点本质上仍是对矩形边界框的建模。ExtremeNet [48]被提出用于定位物体在x和y方向上的极值点[29]，并通过真实掩码标注进行监督。总体而言，自下而上的检测器受益于更小的假设空间（例如，CornerNet和ExtremeNet都检测二维点而非直接检测四维边界框）以及潜在更精细的定位能力。然而，它们也存在局限性，例如依赖手工设计的聚类或后处理步骤来组合完整的物体。

jects from the detected points.

Similar to these bottom-up works, *RepPoints* is also a flexible object representation. However, the representation is constructed in a top-down manner, without the need for handcrafted clustering steps. RepPoints can automatically learn extreme points and key semantic points without supervision beyond ground-truth bounding boxes, unlike ExtremeNet [48] where additional mask supervision is required.

Deformation modeling in object recognition. One of the most fundamental challenges for visual recognition is to recognize objects with various geometric variations. To effectively model such variations, a possible solution is to make use of bottom-up composition of low-level components. Representative detectors along this direction include DPM [9] and Poselet [1]. An alternative is to implicitly model the transformations in a top-down manner, where a lightweight neural network block is applied on input features, either globally [18] or locally [4].

RepPoints is inspired by these works, especially the top-down deformation modeling approach [4]. The main difference is that we aim at developing a flexible object representation for accurate *geometric localization* in addition to semantic feature extraction. In contrast, both the deformable convolution and deformable RoI pooling methods are designed to improve feature extraction only. The inability of deformable RoI pooling to learn accurate geometric localization is examined in Section 4 and appendix. In this sense, we expand the usage of adaptive sample points in previous geometric modeling methods [18, 4] to include finer localization of objects.

3. The RepPoints Representation

We first review the bounding box representation and its use within multi-stage object detectors. This is followed by a description of RepPoints and its differences from bounding boxes.

3.1. Bounding Box Representation

The bounding box is a 4-d representation encoding the spatial location of an object, $\mathcal{B} = (x, y, w, h)$, with x, y denoting the center points and w, h denoting the width and height. Due to its simplicity and convenience in use, modern object detectors heavily rely on bounding boxes for representing objects at various stages of the detection pipeline.

Review of Multi-Stage Object Detectors The best performing object detectors usually follow a multi-stage recognition paradigm, where object localization is refined stage by stage. The role of the object representation through the steps of this pipeline is as follows:

$$\begin{aligned} \text{bbox anchors} &\xrightarrow{\text{bbox reg.}} \text{bbox proposals (S1)} \\ &\xrightarrow{\text{bbox reg.}} \text{bbox proposals (S2)} \\ &\dots \\ &\xrightarrow{\text{bbox reg.}} \text{bbox object targets} \end{aligned} \tag{1}$$

At the beginning, multiple anchors are hypothesized to cover a range of bounding box scales and aspect ratios. In general, high coverage is obtained through dense anchors over the large 4-d hypothesis space. For instance, 45 anchors per location are utilized in RetinaNet [25].

For an anchor, the image feature at its center point is adopted as the object feature, which is then used to produce a confidence score about whether the anchor is an object or not, as well as the refined bounding box by a bounding box regression process. The refined bounding box is denoted as "bbox proposals (S1)".

In the second stage, a refined object feature is extracted from the refined bounding box proposal, usually by RoI-pooling [10] or RoI-Align [13]. For the two-stage framework [33], the refined feature will produce the final bounding box target by bounding box regression. For the multi-stage approach [2], the refined feature is used to generate intermediate refined bounding box proposals (S2), also by bounding box regression. This step can be iterated multiple times before producing the final bounding box target.

In this framework, bounding box regression plays a central role in progressively refining object localization and object features. We formulate the process of bounding box regression in the following paragraph.

Bounding Box Regression Conventionally, a 4-d regression vector $(\Delta x_p, \Delta y_p, \Delta w_p, \Delta h_p)$ is predicted to map the current bounding box proposal $\mathcal{B}_p = (x_p, y_p, w_p, h_p)$ into a refined bounding box $\mathcal{B}_r$, where

$$\mathcal{B}_r = (x_p + w_p\Delta x_p, y_p + h_p\Delta y_p, w_p e^{\Delta w_p}, h_p e^{\Delta h_p}). \tag{2}$$

Given the ground truth bounding box of an object $\mathcal{B}_t = (x_t, y_t, w_t, h_t)$, the goal of bounding box regression is to have $\mathcal{B}_r$ and $\mathcal{B}_t$ as close as possible. Specifically, in the training of an object detector, we use the distance between the predicted 4-d regression vector and the expected 4-d regression vector $\hat{\mathcal{F}}(\mathcal{B}_p, \mathcal{B}_t)$ as the learning target, using a smooth l_1 loss:

$$\hat{\mathcal{F}}(\mathcal{B}_p, \mathcal{B}_t) = (\frac{x_t - x_p}{w_p}, \frac{y_t - y_p}{h_p}, \log\frac{w_t}{w_p}, \log\frac{h_t}{h_p}). \tag{3}$$

This bounding box regression process is widely used in existing object detection methods. It performs well in practice when the required refinement is small, but it tends to perform poorly when there is large distance between the initial representation and the target. Another issue lies in the

从检测到的点中提取物体。

与这些自下而上的工作类似，*RepPoints*同样是一种灵活的对象表示方法。然而，该表示是以自上而下的方式构建的，无需手工设计的聚类步骤。RepPoints能够自动学习极值点和关键语义点，仅需真实标注框作为监督，无需额外标注信息——这与需要额外掩码监督的ExtremeNet[48]不同。

物体识别中的形变建模。视觉识别最根本的挑战之一是如何识别具有各种几何变化的物体。为有效建模此类变化，一种可能的解决方案是利用低级组件的自底向上组合。沿此方向的代表性检测器包括DPM[9]和Poselet[1]。另一种方法是以自顶向下的方式隐式建模变换，即在输入特征上应用轻量级神经网络模块，这种方式既可全局应用[18]也可局部应用[4]。

RepPoints 受到这些工作的启发，尤其是自上而下的变形建模方法[4]。主要区别在于，除了语义特征提取外，我们还致力于开发一种灵活的对象表示以实现精确的*geometric localization*。相比之下，可变形卷积和可变形RoI池化方法都仅旨在改进特征提取。第4节和附录中探讨了可变形RoI池化在学习精确几何定位方面的不足。从这个意义上说，我们将先前几何建模方法[18, 4]中自适应采样点的应用扩展到包含更精细的对象定位。

3. RepPoints表示法

我们首先回顾边界框表示及其在多阶段目标检测器中的应用，随后介绍RepPoints及其与边界框的区别。

3.1. 边界框表示

边界框是一种4维表示，用于编码物体的空间位置，$\mathcal{B} = (x, y, w, h)$，其中$x, y$表示中心点，$w, h$表示宽度和高度。由于其简单性和使用便利性，现代物体检测器在检测流程的各个阶段都严重依赖边界框来表示物体。

多阶段目标检测器综述 性能最佳的目标检测器通常遵循多阶段识别范式，其中目标定位通过逐阶段优化逐步精细化。目标表征在该流程的步骤如下：

$$
\begin{aligned}
\text{bbox anchors} &\xrightarrow{\text{bbox reg.}} \text{bbox proposals (S1)} \\
&\xrightarrow{\text{bbox reg.}} \text{bbox proposals (S2)} \\
&\ \ \cdots \\
&\xrightarrow{\text{bbox reg.}} \text{bbox object targets}
\end{aligned} \tag{1}
$$

起初，假设多个锚框以覆盖不同尺度和长宽比的边界框范围。通常，通过在庞大的4维假设空间中使用密集锚框，可以获得高覆盖率。例如，RetinaNet [25] 中每个位置使用了45个锚框。

对于锚点，我们采用其中心点的图像特征作为物体特征，随后该特征被用于生成一个置信度分数，以判断该锚点是否为物体，并通过边界框回归过程生成精细化的边界框。精细化后的边界框被标记为“边界框提议（S1）”。

在第二阶段，从精炼的边界框提议中提取出精炼的对象特征，通常通过RoI池化[10]或RoI对齐[13]实现。对于两阶段框架[33]，精炼特征将通过边界框回归生成最终的边界框目标。对于多阶段方法[2]，精炼特征则用于生成中间的精炼边界框提议（S2），同样通过边界框回归完成。该步骤可在生成最终边界框目标前多次迭代进行。

在此框架中，边界框回归在逐步优化目标定位与目标特征方面起着核心作用。我们将在下文中阐述边界框回归的过程。

边界框回归传统上，预测一个4维回归向量（{v*}）以将当前边界框提议 {v*} 映射到一个精炼的边界框 {v*}，其中

$$\mathcal{B}_r = (x_p + w_p\Delta x_p, y_p + h_p\Delta y_p, w_p e^{\Delta w_p}, h_p e^{\Delta h_p}). \tag{2}$$

给定物体的真实边界框 $\mathcal{B}_t = (x_t, y_t, w_t, h_t)$，边界框回归的目标是使 $\mathcal{B}_r$ 和 $\mathcal{B}_t$ 尽可能接近。具体而言，在目标检测器的训练中，我们使用预测的4维回归向量与期望的4维回归向量$\hat{\mathcal{F}}(\mathcal{B}_p, \mathcal{B}_t)$ 之间的距离作为学习目标，并采用平滑的 l_1 损失函数：

$$\hat{\mathcal{F}}(\mathcal{B}_p, \mathcal{B}_t) = (\frac{x_t - x_p}{w_p}, \frac{y_t - y_p}{h_p}, \log\frac{w_t}{w_p}, \log\frac{h_t}{h_p}). \tag{3}$$

该边界框回归过程在现有目标检测方法中被广泛使用。当所需修正较小时，它在实践中表现良好，但当初始表示与目标之间存在较大距离时，其性能往往较差。另一个问题在于

scale difference between $\Delta x, \Delta y$ and $\Delta w, \Delta h$, which requires tuning of their loss weights for optimal performance.

3.2. RepPoints

As previously discussed, the 4-d bounding box is a coarse representation of object location. The bounding box representation considers only the rectangular spatial scope of an object, and does not account for shape and pose and the positions of semantically important local areas, which could be used toward finer localization and better object feature extraction.

To overcome the above limitations, *RepPoints* instead models a set of adaptive sample points:

$$\mathcal{R} = \{(x_k, y_k)\}_{k=1}^{n}, \tag{4}$$

where n is the total number of sample points used in the representation. In our work, n is set to 9 by default.

RepPoints refinement Progressively refining the bounding box localization and feature extraction is important for the success of multi-stage object detection methods. For RepPoints, the refinement can be expressed simply as

$$\mathcal{R}_r = \{(x_k + \Delta x_k, y_k + \Delta y_k)\}_{k=1}^{n}, \tag{5}$$

where $\{(\Delta x_k, \Delta y_k)\}_{k=1}^{n}$ are the predicted offsets of the new sample points with respect to the old ones. We note that this refinement does not face the problem of scale differences among the bounding box regression parameters, since the offsets are at the same scale in the refinement process of *RepPoints*.

Converting RepPoints to bounding box To take advantage of bounding box annotations in the training of RepPoints, as well as to evaluate RepPoint-based object detectors, a method is needed for converting RepPoints into a bounding box. We perform this conversion using a predefined converting function $\mathcal{T} : \mathcal{R}_P \rightarrow \mathcal{B}_P$, where $\mathcal{R}_P$ denotes the RepPoints for object P and $\mathcal{T}(\mathcal{R}_P)$ represents a *pseudo box*.

Three converting functions are considered for this purpose:

- $\mathcal{T} = \mathcal{T}_1$**: Min-max function.** Min-max operation over both axes are performed over the *RepPoints* to determine $\mathcal{B}_p$, equivalent to the bounding box over the sample points.

- $\mathcal{T} = \mathcal{T}_2$**: Partial min-max function.** Min-max operation over a subset of the sample points is performed over both axes to obtain the rectangular box $\mathcal{B}_p$.

- $\mathcal{T} = \mathcal{T}_3$**: Moment-based function.** The mean value and the standard deviation of the RepPoints is used to compute the center point and scale of the rectangular box $\mathcal{B}_p$, where the scale is multiplied by globally-shared learnable multipliers λ_x and λ_y.

These functions are all differentiable, enabling end-to-end learning when inserted into an object detection system. In our experiments, we found them to work comparably well.

Learning RepPoints The learning of RepPoints is driven by both an object localization loss and an object recognition loss. To compute the object localization loss, we first convert *RepPoints* into a *pseudo box* using the previously discussed transformation function $\mathcal{T}$. Then, the difference between the converted *pseudo box* and the ground-truth bounding box is computed. In our system, we use the smooth l_1 distance between the top-left and bottom-right points to represent the localization loss. This smooth l_1 distance does not require the tuning of different loss weights as done in computing the distance between bounding box regression vectors (i.e., for $\Delta x, \Delta y$ and $\Delta w, \Delta h$). Figure 4 indicates that when the training is driven by this combination of object localization and object recognition losses, the extreme points and semantic key points of objects are automatically learned ($\mathcal{T}_1$ is used in transforming RepPoints to pseudo box).

4. RPDet: an Anchor Free Detector

We design an anchor-free object detector that utilizes RepPoints in place of bounding boxes as its basic representation. Within a multi-stage pipeline, the object representation evolves as follows:

$$\begin{aligned} \text{object centers} &\xrightarrow{\text{RP refine}} \text{RepPoints proposals (S1)} \\ &\xrightarrow{\text{RP refine}} \text{RepPoints proposals (S2)} \\ &\ldots \\ &\xrightarrow{\text{RP refine}} \text{RepPoints object targets} \end{aligned} \tag{6}$$

Our RepPoints Detector (RPDet) is constructed with two recognition stages based on deformable convolution, as illustrated in Figure 2. Deformable convolution pairs nicely with RepPoints, as its convolutions are computed on an irregularly distributed set of sample points and conversely its recognition feedback can guide training for the positioning of these points. In this section, we present the design of RPDet and discuss its relationship to and differences from existing object detectors.

Center point based initial object representation. While predefined anchors dominate the representation of objects in the initial stage of object detection, we follow YOLO [31] and DenseBox [17] by using center points as the initial representation of objects, which leads to an anchor-free object detector.

An important benefit of the center point representation lies in its much tighter hypothesis space compared to the anchor based counterparts. While anchor based approaches usually rely on a large number of multi-ratio and multi-scale anchors to ensure dense coverage of the large 4-d bounding

$\Delta x, \Delta y$和$\Delta w, \Delta h$之间的尺度差异，这需要调整它们的损失权重以获得最佳性能。

3.2. 代表点

如前所述，4维边界框是物体位置的粗略表示。边界框表示仅考虑物体的矩形空间范围，未考虑形状、姿态以及语义重要局部区域的位置，而这些信息可用于更精细的定位和更好的物体特征提取。

为了克服上述限制，*RepPoints*转而建模一组自适应采样点：

$$\mathcal{R} = \{(x_k, y_k)\}_{k=1}^{n}, \tag{4}$$

其中 n 是表示中使用的样本点总数。在我们的工作中，n 默认设置为 9。

RepPoints 的逐步优化 逐步优化边界框定位和特征提取对于多阶段目标检测方法的成功至关重要。对于RepPoints而言，这种优化可以简单表示为

$$\mathcal{R}_r = \{(x_k + \Delta x_k, y_k + \Delta y_k)\}_{k=1}^{n}, \tag{5}$$

其中$\{(\Delta x_k, \Delta y_k)\}_{k=1}^{n}$是新样本点相对于旧样本点的预测偏移量。我们注意到，这种优化不会面临边界框回归参数间尺度差异的问题，因为在*RepPoints*的优化过程中，偏移量处于同一尺度。

将RepPoints转换为边界框 为了在RepPoints的训练中利用边界框标注，并评估基于RepPoints的目标检测器，需要一种将RepPoints转换为边界框的方法。我们使用预定义的转换函数$\mathcal{T}$: $\mathcal{R}_P \to \mathcal{B}_P$来执行此转换，其中$\mathcal{R}_P$表示目标$P$的RepPoints，$\mathcal{T}(\mathcal{R}_P)$代表一个*pseudo box*。

为此目的，考虑了三种转换函数：

- $\mathcal{T} = \mathcal{T}_1$: 最小-最大函数。在*RepPoints*上沿双轴执行最小-最大运算以确定$\mathcal{B}_p$，这等价于样本点上的边界框。
- $\mathcal{T} = \mathcal{T}_2$: 部分最小-最大函数。在样本点子集上执行最小-最大操作，覆盖两个坐标轴以获得矩形框$\mathcal{B}_p$。
- $\mathcal{T} = \mathcal{T}_3$: 基于矩的函数。利用RepPoints的均值与标准差计算矩形框的中心点与尺度$\mathcal{B}_p$，其中尺度会乘以全局共享的可学习乘数λ_x和λ_y。

这些函数都是可微分的，当它们被插入到目标检测系统中时，能够实现端到端的学习。在我们的实验中，我们发现它们表现相当出色。

学习RepPoints由目标定位损失和目标识别损失共同驱动。为计算目标定位损失，我们首先使用先前讨论的转换函数$\mathcal{T}$将*RepPoints*转换为*pseudo box*，随后计算转换后的*pseudo box*与真实边界框之间的差异。本系统采用左上角与右下角点之间的平滑l_1距离来表示定位损失，这种平滑l_1距离无需像计算边界框回归向量距离时（即针对$\Delta x, \Delta y$和$\Delta w, \Delta h$）那样调整不同损失的权重。图4表明，当训练由目标定位与目标识别损失共同驱动时，物体的极端点和语义关键点会被自动学习（在将RepPoints转换为伪框时使用了$\mathcal{T}_1$）。

4. RPDet：一种无锚框检测器

我们设计了一种无锚框目标检测器，它使用RepPoints代替边界框作为基本表示。在多阶段流程中，目标表示按如下方式演进：

$$\begin{array}{lcl} \text{object centers} & \xrightarrow{\text{RP refine}} & \text{RepPoints proposals (S1)} \\ & \xrightarrow{\text{RP refine}} & \text{RepPoints proposals (S2)} \\ & \ldots & \\ & \xrightarrow{\text{RP refine}} & \text{RepPoints object targets} \end{array} \tag{6}$$

我们的RepPoints检测器（RPDet）基于可变形卷积构建，包含两个识别阶段，如图2所示。可变形卷积与RepPoints能很好地配合，因为其卷积计算在一组不规则分布的采样点上进行，反之其识别反馈也能指导这些点的定位训练。本节将介绍RPDet的设计，并讨论其与现有物体检测器的关联及差异。

基于中心点的初始物体表示。在物体检测的初始阶段，预定义锚点主导着物体的表示，我们遵循YOLO [31]和DenseBox [17]的方法，采用中心点作为物体的初始表示，从而构建了一种无锚框的物体检测器。

中心点表示的一个重要优势在于，与基于锚点的方法相比，其假设空间更为紧凑。基于锚点的方法通常依赖大量多比例、多尺度的锚点来确保对四维边界框空间的密集覆盖。

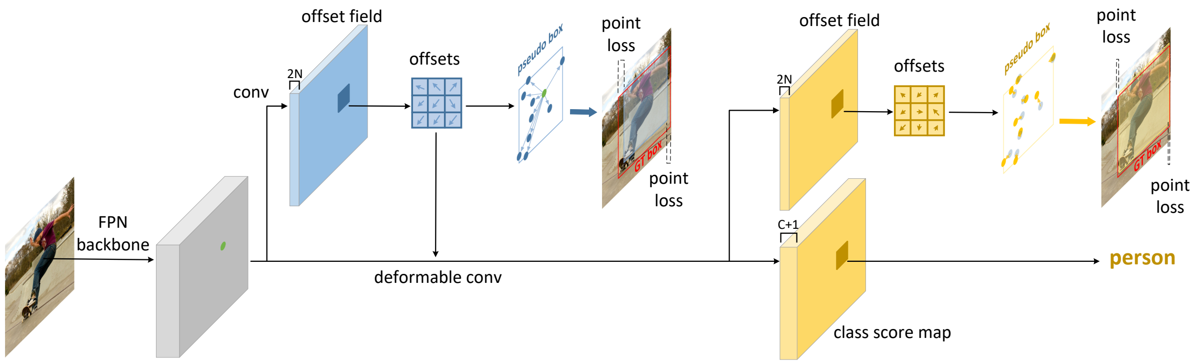

Figure 2. Overview of the proposed RPDet (RepPoints detector). While feature pyramidal networks (FPN) [24] are adopted as the backbone, we only draw the afterwards pipeline of one scale of FPN feature maps for clear illustration. Note all scales of FPN feature maps share the same afterwards network architecture and the same model weights.

box hypothesis space, a center point based approach can more easily cover its 2-d space. In fact, all objects will have center points located within the image.

However, the center point based method also faces the problem of recognition target ambiguity, caused by two different objects locating at the same position in a feature map, which limits its prevalence in modern object detectors. In previous methods [31], this is mainly addressed by producing multiple targets at each position, which faces another issue of vesting ambiguity[1]. In RPDet, we show that this issue can be greatly alleviated by using the FPN structure [24] for the following reasons: first, objects of different scales will be assigned to different image feature levels, which addresses objects of different scales and the same center points locations; second, FPN has a high-resolution feature map for small objects, which also reduces the chance of two objects having centers located at the same feature position. In fact, we observe that only 1.1% of objects in the COCO datasets [26] suffer from the issue of center points located at the same position when FPN is used.

It is worth noting that the center point representation can be viewed as a special RepPoints configuration, where only a single fixed sample point is used, thus maintaining a coherent representation throughout the proposed detection framework.

Utilization of RepPoints. As shown in Figure 2, RepPoints serve as the basic object representation throughout our detection system. Starting from the center points, the first set of *RepPoints* is obtained via regressing offsets over the center points. The learning of these *RepPoints* is driven by two objectives: 1) the top-left and bottom-right points distance loss between the induced *pseudo box* and the ground-truth bounding box; 2) the object recognition loss of the subsequent stage. As illustrated in Figure 4, extreme

[1]If the center points of multiple ground truth objects are located at a same feature map position, only one randomly chosen ground truth object is assigned to be the target of this position.

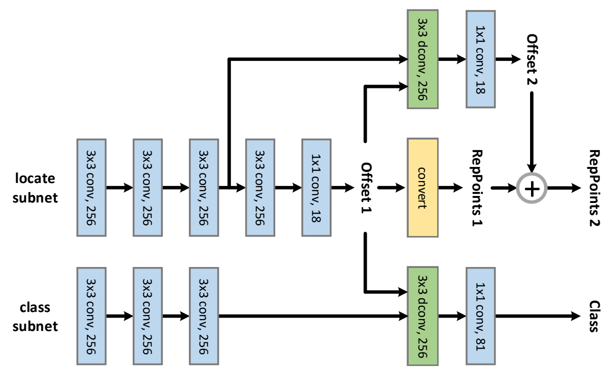

Figure 3. The head architecture of RPDet.

and key points are automatically learned. The second set of *RepPoints* represents the final object localization, which is refined from the first set of *RepPoints* by Eq. (5). Driven by the points distance loss alone, this second set of *RepPoints* aims to learn finer object localization.

Relation to deformable RoI pooling [4]. As mentioned in Section 2, deformable RoI pooling plays a different role in object detection compared to the proposed *RepPoints*. Basically, *RepPoints* is a geometric representation of objects, reflecting more accurate semantic localization, while deformable RoI pooling is geared towards learning stronger appearance features of objects. In fact, deformable RoI pooling cannot learn sample points representing accurate localization of objects (please see the appendix for a proof).

We also note that deformable RoI pooling can be complementary to RepPoints, as indicated in Table 6.

Backbone and head architectures Our FPN backbone follows [25], which produces 5 feature pyramid levels from stage 3 (downsampling ratio of 8) to stage 7 (downsampling ratio of 128).

The head architecture is illustrated in Figure 3. There are two *non-shared* subnets, aiming at localization (RepPoints generation) and classification, respectively. The localization subnet first apply three 256-d 3×3 conv layers, fol-

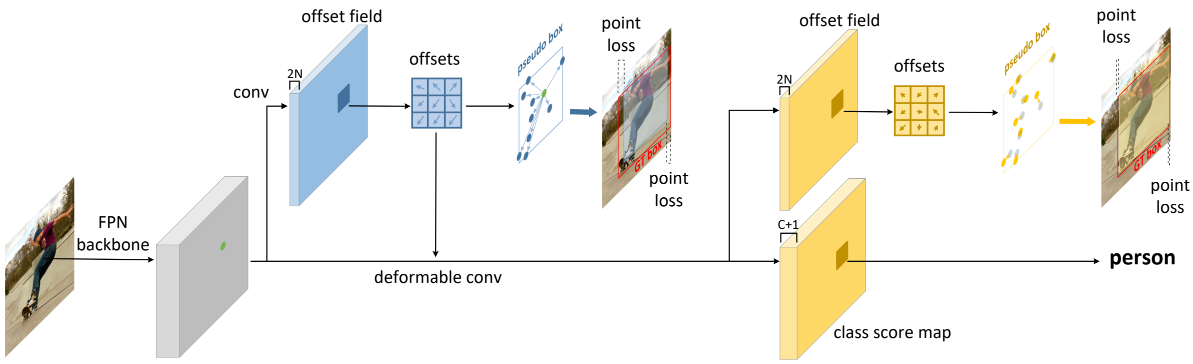

图2. 所提出的RPDet（RepPoints检测器）概述。虽然采用特征金字塔网络（FPN）[24]作为骨干网络，但为清晰说明，我们仅绘制了单尺度FPN特征图的后继流程。请注意，所有尺度的FPN特征图共享相同的后继网络架构和模型权重。

在框假设空间中，基于中心点的方法能更轻松地覆盖其二维空间。事实上，所有物体的中心点都将位于图像内部。

然而，基于中心点的方法也面临着识别目标模糊的问题，这是由于两个不同物体在特征图中位于同一位置所导致的，这限制了其在现代物体检测器中的普及。在先前的方法[31]中，主要通过在每个位置生成多个目标来解决这一问题，但这又带来了归属模糊[1]的问题。在RPDet中，我们展示了通过使用FPN结构[24]可以极大地缓解这一问题，原因如下：首先，不同尺度的物体会被分配到不同的图像特征层级，这解决了不同尺度物体具有相同中心点位置的问题；其次，FPN为小物体提供了高分辨率的特征图，这也降低了两物体中心位于同一特征位置的可能性。事实上，我们观察到，在使用FPN时，COCO数据集[26]中仅有1.1%的物体存在中心点位于同一位置的问题。

值得注意的是，中心点表示可视为一种特殊的RepPoints配置，其中仅使用单个固定采样点，从而在整个检测框架中保持连贯的表示形式。

RepPoints的利用。如图2所示，RepPoints作为我们检测系统中的基本对象表示。从中心点出发，第一组{v*}通过回归中心点上的偏移量获得。这些{v*}的学习由两个目标驱动：1）诱导出的{v*}与真实边界框之间左上和右下点的距离损失；2）后续阶段的对象识别损失。如图4所示，极端

[1]If the center points of multiple ground truth objects are located at a same feature map position, only one randomly chosen ground truth object is assigned to be the target of this position.

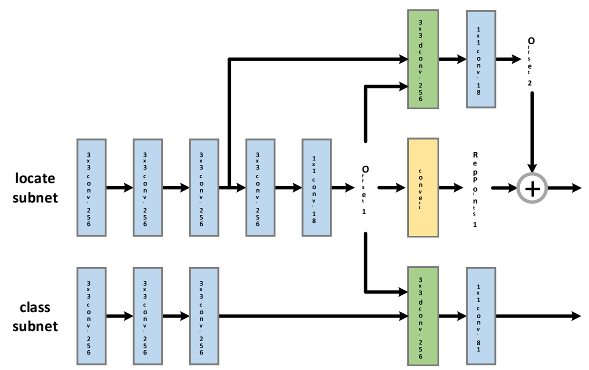

图3. RPDet的头部架构。

并且关键点被自动学习。第二组*RepPoints*代表最终的对象定位，它通过方程(5)从第一组*RepPoints*中细化而来。仅由点距离损失驱动，这第二组*RepPoints*旨在学习更精细的对象定位。

与可变形RoI池化[4]的关系。如第2节所述，可变形RoI池化在目标检测中扮演的角色与提出的*RepPoints*不同。本质上，*RepPoints*是目标的几何表征，反映了更精确的语义定位，而可变形RoI池化则旨在学习目标更强的外观特征。实际上，可变形RoI池化无法学习到表示目标精确定位的采样点（证明详见附录）。

我们还注意到，可变形RoI池化可以作为RepPoints的补充，如表6所示。

骨干网络与头部架构 我们的FPN骨干网络遵循[25]的设计，从第3阶段（下采样率为8）到第7阶段（下采样率为128）共生成5个特征金字塔层级。

头部架构如图3所示。包含两个*non-shared*子网络，分别用于定位（RepPoints生成）和分类。定位子网络首先应用三个256维3×3卷积层，后接——

lowed by two successive small networks to compute offsets for the two sets of *RepPoints*. The classification subnetwork also apply three 256-d 3×3 conv layers, followed by a 256-d 3×3 deformable conv layer with its input offset field shared with that of the first deformable conv layer in the localization subnetwork. The group normalization layer is applied after each of the first three 256-d 3×3 conv layers in both subnets.

Note although our approach utilizes two stages of localization, it is even more efficient than the one stage RetinaNet [25] detector (210.9 vs. 234.5 GFLOPS using ResNet-50). The additional localization stage introduces little overhead due to layer sharing. The anchor-free design reduces the burden of the final classification layer which leads to a slight reduction in computation.

Localization/class target assignment. There are two localization stages: generating the first set of RepPoints by refining from the object center point hypothesis (feature map bins); generating the second set of RepPoints by refining from the first RepPoints set. For both stages, only *positive* object hypothesis are assigned with localization (RepPoints) targets in training. For the first localization stage, a feature map bin is *positive* if 1) the pyramidal level of this feature map bin equals the log scale of a ground-truth object $s(B) = \lfloor log_2(\sqrt{w_B h_B}/4) \rfloor$; 2) the projection of this ground-truth object's center point locate within this feature map bin. For the second localization stage, the first RepPoints is *positive* if its induced pseudo box has sufficient overlap with a ground-truth object that their intersection-over-union is larger than 0.5.

Classification is conducted on the first set of RepPoints only. The classification assignment criterion follows [25] that IoU (between the induced pseudo box and ground-truth bounding box) larger than 0.5 indicates *positive*, smaller than 0.4 indicates background, and otherwise ignored. Focal loss is employed for classification training [25].

5. Experiments

5.1. Experimental Settings

We present experimental results of our proposed RPDet framework on the MS-COCO [26] detection benchmark, which contains 118k images for training, 5k images for validation (`minival`) and 20k images for testing (`test-dev`). All the ablation studies are conducted on `minival` with ResNet-50 [15], if not otherwise specified. The state-of-the-art comparison is reported on `test-dev` in Table 7.

Our detector is trained with synchronized stochastic gradient descent (SGD) over 4 GPUs with a total of 8 images per minibatch (2 images per GPU). The ImageNet [6] pre-trained model was used for initialization. Our learning rate schedule follows the '1x' setting [12]. Random horizontal image flipping is adopted in training. In inference, NMS ($\sigma = 0.5$) is employed to post-process the results, following [25].

Representation	Backbone	AP	AP_{50}	AP_{75}
Bounding box	ResNet-50	36.2	57.3	39.8
RepPoints (ours)	ResNet-50	**38.3**	**60.0**	**41.1**
Bounding box	ResNet-101	38.4	59.9	42.4
RepPoints (ours)	ResNet-101	**40.4**	**62.0**	**43.6**

Table 1. Comparison of the RepPoints and bounding box representations in object detection. The network structures are the same except for processing the given object representation.

Representation	Supervision loc.	Supervision rec.	AP	AP_{50}	AP_{75}
bounding box	✓		36.2	57.3	39.8
	✓	✓	36.2	57.5	39.8
RepPoints		✓	33.8	54.3	35.8
	✓		37.6	59.4	40.4
	✓	✓	**38.3**	**60.0**	**41.1**

Table 2. Ablation of the supervision sources, for both bounding box and RepPoints based object detection. "loc." indicates the object localization loss. "rec." indicates the object recognition loss from the next detection stage.

other repr.	init repr.	AP	AP_{50}	AP_{75}
bounding box	single anchor	36.2	57.3	39.8
	center point	**37.3**	**58.0**	**40.0**
RepPoints	single anchor	36.9	58.2	39.7
	center point	**38.3**	**60.0**	**41.1**

Table 3. Comparison of using a single anchor and a center point as the initial object representation ("init repr."). "other repr." indicates representation method for object proposals and final targets.

method	backbone	# anchors per scale	AP
RetinaNet [25]	ResNet-50	3×3	35.7
FPN-RoIAlign [24]	ResNet-50	3×1	36.7
YOLO-like	ResNet-50	-	33.9
RPDet (ours)	ResNet-50	-	**38.3**
RetinaNet [25]	ResNet-101	3×3	37.8
FPN-RoIAlign [24]	ResNet-101	3×1	39.4
YOLO-like	ResNet-101	-	36.3
RPDet (ours)	ResNet-101	-	**40.4**

Table 4. Comparison of the proposed method (RPDet) with anchor-based methods (RetinaNet, FPN-RoIAlign) and an anchor-free method (YOLO-like). The YOLO-like method is adapted from the YOLOv1 method [31] by additionally introducing FPN [24], GN [42] and focal loss [25] for better accuracy.

随后是两个连续的小型网络，用于计算两组*RepPoints*的偏移量。分类子网络同样应用三个256维3×3卷积层，接着是一个256维3×3可变形卷积层，其输入偏移场与定位子网络中第一个可变形卷积层的偏移场共享。在两个子网络中，前三个256维3×3卷积层之后均应用了组归一化层。

请注意，尽管我们的方法采用了两个阶段的定位，但它甚至比单阶段的RetinaNet [25] 检测器更高效（使用ResNet-50时分别为210.9 vs. 234.5 GFLOPS）。由于层共享，额外的定位阶段引入的开销很小。无锚框设计减轻了最终分类层的负担，从而略微减少了计算量。

定位/分类目标分配。存在两个定位阶段：首先生成第一组RepPoints，通过从物体中心点假设（特征图单元）进行细化；然后生成第二组RepPoints，通过从第一组RepPoints进行细化。在这两个阶段中，训练时仅对*posi- tive*物体假设分配定位（RepPoints）目标。对于第一定位阶段，一个特征图单元被*positive*的条件是：1）该特征图单元的金字塔层级等于真实物体$s(B) = \lfloor log_2(\sqrt{w_B h_B}/4) \rfloor$的对数尺度；2）该真实物体中心点的投影位于此特征图单元内。对于第二定位阶段，第一组RepPoints被*positive*的条件是：其诱导出的伪框与某个真实物体有足够重叠，即它们的交并比大于0.5。

分类仅在第一组RepPoints上进行。分类分配标准遵循[25]，即诱导出的伪框与真实边界框之间的IoU大于0.5表示*positive*，小于0.4表示背景，其余情况则忽略。分类训练采用焦点损失[25]。

5. 实验

5.1. 实验设置

我们在MS-COCO [26]检测基准上展示了所提出的RPDet框架的实验结果，该基准包含118k张训练图像、5k张验证图像（minival）和20k张测试图像（test-dev）。除非另有说明，所有消融研究均使用ResNet-50 [15]在minival上进行。表7中报告了在test-dev上的最先进性能对比。

我们的检测器使用同步随机梯度下降（SGD）在4个GPU上进行训练，每个小批次共包含8张图像（每个GPU处理2张图像）。初始化时采用了ImageNet [6]的预训练模型。我们的学习率调度遵循“1x”设置 [12]。随机水平

Representation	Backbone	AP	AP_{50}	AP_{75}
Bounding box	ResNet-50	36.2	57.3	39.8
RepPoints (ours)	ResNet-50	**38.3**	**60.0**	**41.1**
Bounding box	ResNet-101	38.4	59.9	42.4
RepPoints (ours)	ResNet-101	**40.4**	**62.0**	**43.6**

表1. 目标检测中RepPoints与边界框表示方法的比较。除处理给定目标表示方式外，网络结构保持相同。

Representation	Supervision		AP	AP_{50}	AP_{75}
	loc.	rec.			
bounding box	✓		36.2	57.3	39.8
	✓	✓	36.2	57.5	39.8
RepPoints		✓	33.8	54.3	35.8
	✓		37.6	59.4	40.4
	✓	✓	**38.3**	**60.0**	**41.1**

表2. 基于边界框和RepPoints的目标检测中监督来源的消融实验。“loc.”表示目标定位损失。“rec.”表示来自下一检测阶段的目标识别损失。

other repr.	init repr.	AP	AP_{50}	AP_{75}
bounding box	single anchor	36.2	57.3	39.8
	center point	**37.3**	**58.0**	**40.0**
RepPoints	single anchor	36.9	58.2	39.7
	center point	**38.3**	**60.0**	**41.1**

表3. 使用单个锚点与中心点作为初始对象表示（“初始表示”）的对比。“其他表示”指代对象提议和最终目标的表示方法。

method	backbone	# anchors per scale	AP
RetinaNet [25]	ResNet-50	3×3	35.7
FPN-RoIAlign [24]	ResNet-50	3×1	36.7
YOLO-like	ResNet-50	-	33.9
RPDet (ours)	ResNet-50	-	**38.3**
RetinaNet [25]	ResNet-101	3×3	37.8
FPN-RoIAlign [24]	ResNet-101	3×1	39.4
YOLO-like	ResNet-101	-	36.3
RPDet (ours)	ResNet-101	-	**40.4**

表4. 所提方法（RPDet）与基于锚框的方法（RetinaNet、FPN-RoIAlign）以及无锚框方法（类YOLO方法）的对比。类YOLO方法基于YOLOv1方法[31]改进，额外引入了FPN[24]、GN[42]和焦点损失[25]以提升精度。

在训练中采用了水平图像翻转。在推理过程中，按照[25]的方法，使用NMS（{v*}0.5）对结果进行后处理。

Figure 4. Visualization of the learnt RepPoints and the induced bounding boxes on several examples from the COCO [26] minival set (using the pseudo box converting function $\mathcal{T}_1$). In general, the learned RepPoints are located on extreme or semantic keypoints of objects.

pseudo box converting function	AP	AP_{50}	AP_{75}
$\mathcal{T} = \mathcal{T}_1$: min-max	38.2	59.7	40.7
$\mathcal{T} = \mathcal{T}_2$: partial min-max	38.1	59.6	40.5
$\mathcal{T} = \mathcal{T}_3$: moment-based	38.3	60.0	41.1

Table 5. Comparison of different transformation functions from RepPoints to pseudo box, $\mathcal{T}$.

representation method	w. dpool	AP	AP_{50}	AP_{75}
bounding box		36.2	57.3	39.8
	✓	36.9	58.0	41.0
RepPoints		38.3	60.0	41.1
	✓	**39.1**	**60.6**	**42.4**

Table 6. The effect of applying the deformable RoI pooling layer [4] on the proposals of the first stages (see Eq. (1) and Eq. (6)). The deformable RoI pooling layer can boost both the methods using bounding boxes and RepPoints, respectively.

5.2. Ablation Study

RepPoints vs. bounding box. To demonstrate the effectiveness of the proposed *RepPoints*, we compare the proposed RPDet to a baseline detector where RepPoints are all replaced by the regular bounding box representation.

Baseline detector based on bounding box representations. A single anchor with scale of 4 and aspect ratio of $1:1$ is adopted for the initial object representation, where the anchor box is *positive* if the IoU with a ground-truth object is larger than 0.5. The two sets of RepPoints are replaced by bounding box representation, where the geometric refinement is achieved by the standard bounding box regression method, and the feature extraction is replaced by the RoIAlign [13] method using 3×3 grid points[2]. All other settings are the same as in the proposed RPDet method.

Table 1 shows the comparison of two detectors. While the bounding box based method achieves 36.2 mAP, indicating a strong baseline detector, the change of object representation from bounding box to RepPoints brings a +2.1 mAP improvement using ResNet-50 [15] and a +2.0 mAP improvement using a ResNet-101 [15] backbone, demonstrating the advantage of the RepPoints representation over bounding boxes for object detection.

Supervision source for RepPoints learning. RPDet uses both an object localization loss and an object recognition loss (gradient from later recognition) to drive the learning of the first set of RepPoints, which represents the object proposals of the first stage. Table 2 ablates the use of these two supervision sources in the learning of the object representations. As mentioned before, describing the geometric localization of objects is an important duty of a representation method. Without the object localization loss, it is hard for a representation method to accomplish this duty, as it results in significant performance degradation of the object detectors. For RepPoints, we observe a huge drop of 4.5 mAP by removing the object localization supervision, showing the importance of describing the geometric localization for an object representation method.

Table 2 also demonstrates the benefit of inluding the object recognition loss in learning RepPoints (+0.7 mAP). The use of the object recognition loss can drive the RepPoints to locate themselves at semantically meaningful positions on an object, which leads to fine-grained localization and improves object feature extraction for the following recognition stage. Note that the object recognition feedback cannot benefit object detection with the bounding box representation (see the first block in Table 2), further demonstrating the advantage of RepPoints in flexible object representation.

Anchor-free vs. anchor-based. We first compare the center point based method (a special RepPoints configuration) and the prevalent anchor based method in representing initial object hypotheses, in Table 3. For both detectors us-

[2]It can be also implemented by a deformable convolution operator with an unlearnable input offset field induced by the 3×3 grid points.

图4. 在COCO [26] 小型验证集的若干示例上，学习到的RepPoints及由其导出的边界框可视化（使用伪框转换函数 $\mathcal{T}_1$）。总体而言，学习到的RepPoints通常位于物体的极值点或语义关键点上。

pseudo box converting function	AP	AP_{50}	AP_{75}
$\mathcal{T} = \mathcal{T}_1$: min-max	38.2	59.7	40.7
$\mathcal{T} = \mathcal{T}_2$: partial min-max	38.1	59.6	40.5
$\mathcal{T} = \mathcal{T}_3$: moment-based	38.3	60.0	41.1

表5. 从RepPoints到伪框的不同转换函数比较，$\mathcal{T}$。

representation method	w. dpool	AP	AP_{50}	AP_{75}
bounding box		36.2	57.3	39.8
	✓	36.9	58.0	41.0
RepPoints		38.3	60.0	41.1
	✓	**39.1**	**60.6**	**42.4**

表6. 在首阶段提议上应用可变形RoI池化层[4]的效果（参见公式(1)与公式(6)）。可变形RoI池化层能分别提升使用边界框和RepPoints两种方法的性能。

5.2. 消融研究

RepPoints与边界框对比。为了展示所提出的*RepPoints*的有效性，我们将提出的RPDet与一个基线检测器进行比较，其中RepPoints全部被常规的边界框表示所取代。

Baseline detector based on bounding box representations. 初始目标表示采用一个尺度为4、宽高比为1:1的单一锚点，若其与真实目标的交并比大于0.5，则锚框为*positive*。两组RepPoints被替换为边界框表示，其中几何细化通过标准边界框回归方法实现，特征提取则采用RoIAlign[13]方法，使用3×3网格点[2]。其余所有设置均与所提出的RPDet方法保持一致。

表1展示了两种检测器的对比。基于边界框的方法达到了36.2 mAP，表明其作为基线检测器的强大性能，而将物体表示从边界框改为RepPoints后，在使用ResNet-50 [15] 时带来了+2.1 mAP的提升，在使用ResNet-101 [15] 骨干网络时带来了+2.0 mAP的提升，这证明了在物体检测任务中RepPoints表示相较于边界框的优势。

RepPoints学习的监督来源。RPDet利用目标定位损失和目标识别损失（来自后续识别的梯度）共同驱动学习过程。

[2]It can be also implemented by a deformable convolution operator with an unlearnable input offset field induced by the 3×3 grid points.

第一组RepPoints，代表了第一阶段的物体提议。表2消融了这两种监督来源在物体表示学习中的使用。如前所述，描述物体的几何定位是表示方法的重要职责。没有物体定位损失，表示方法很难完成这一职责，因为这会导致物体检测器的性能显著下降。对于RepPoints，我们观察到移除物体定位监督后mAP大幅下降了4.5，这表明描述几何定位对于物体表示方法的重要性。

表2同样展示了在学习RepPoints时加入物体识别损失的好处（{v*}0.7 mAP）。使用物体识别损失能够驱动RepPoints定位到物体上具有语义意义的位置，从而实现细粒度的定位，并提升后续识别阶段的物体特征提取效果。值得注意的是，物体识别反馈无法使采用边界框表示的目标检测方法受益（见表2第一栏），这进一步证明了RepPoints在灵活物体表示方面的优势。

无锚框与基于锚框的方法。我们首先在表3中比较了基于中心点的方法（一种特殊的RepPoints配置）和主流的基于锚框的方法在表示初始目标假设方面的表现。对于两种检测器使

	Backbone	Anchor-Free	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Faster R-CNN w. FPN [24]	ResNet-101		36.2	59.1	39.0	18.2	39.0	48.2
RefineDet [46]	ResNet-101		36.4	57.5	39.5	16.6	39.9	51.4
RetinaNet [25]	ResNet-101		39.1	59.1	42.3	21.8	42.7	50.2
Deep Regionlets [44]	ResNet-101		39.3	59.8	-	21.7	43.7	50.9
Mask R-CNN [13]	ResNeXt-101		39.8	62.3	43.4	22.1	43.2	51.2
FSAF [50]	ResNet-101		40.9	61.5	44.0	24.0	44.2	51.3
Cascade R-CNN [2]	ResNet-101		42.8	62.1	46.3	23.7	45.5	55.2
CornerNet [21]	Hourglass-104	✓	40.5	56.5	43.1	19.4	42.7	53.9
ExtremeNet [48]	Hourglass-104	✓	40.1	55.3	43.2	20.3	43.2	53.1
RPDet	ResNet-101	✓	41.0	62.9	44.3	23.6	44.1	51.7
RPDet	ResNet-101-DCN	✓	**42.8**	**65.0**	**46.3**	**24.9**	**46.2**	**54.7**
RPDet (ms train)	ResNet-101-DCN	✓	**45.0**	**66.1**	**49.0**	**26.6**	**48.6**	**57.5**
RPDet (ms train & ms test)	ResNet-101-DCN	✓	**46.5**	**67.4**	**50.9**	**30.3**	**49.7**	**57.1**

Table 7. Comparison of the proposed RPDet with the previous state-of-the-art detectors on COCO [26] `test-dev`. The proposed RPDet can achieve 46.5 AP, significantly surpasses all other detectors. Also note the AP_{50} of RPDet is relatively high, which is believed as a better metric in many applications [32]. "ms" indicates multi-scale.

ing bounding boxes and RepPoints, the center point based method surpass the anchor based method by +1.1 mAP and +1.4 mAP, respectively, likely because of its better coverage of ground-truth objects.

We also compare the proposed anchor-free detector based on RepPoints to RetinaNet [25] (a popular one-stage anchor-based method), FPN [24] with RoIAlign (a popular two-stage anchor-based method) [13], and a YOLO-like detector which is adapted from the anchor-free method of YOLOv1 [31], in Table 4. The proposed method outperforms both RetinaNet [25] and the FPN [24] methods, which utilize multiple anchors per scale and sophisticated anchor configurations (FPN). The proposed method also significantly surpasses another anchor-free method (the YOLO-like detector) by +4.4 mAP and +4.1 mAP, respectively, probably due to the flexible RepPoints representation and its effective refinement.

Converting RepPoints to pseudo box. Table 5 shows that different instantiations of the transformation functions $\mathcal{T}$ presented in Section 3.2 work comparably well.

RepPoints act complementary to deformable RoI pooling [4]. Table 6 shows the effect of applying the deformable RoI pooling layer [4] to both bounding box proposals and RepPoints proposals. While applying the deformable RoI pooling layer to bounding box proposals brings a +0.7 mAP gain, applying it to RepPoints proposals also brings a +0.8 mAP gain, implying that the roles of deformable RoI pooling and the proposed RepPoints are complementary.

5.3. RepPoints Visualization

In Figure 4, we visualize the learned *RepPoints* and the corresponding detection results on several examples from the COCO [26] `minival` set. It can be observed that *RepPoints* tend to be located at extreme points or key semantic points of objects. These point distributed over objects are automatically learned without explicit supervision. The visualized results also indicate that the proposed RPDet, implemented here with the min-max transformation function, can effectively detect tiny objects.

5.4. State-of-the-art Comparison

We compare RPDet with state-of-the-art detectors on `test-dev`. Table 7 shows the results. Without any bells and whistles, RPDet achieves 42.8 AP on COCO benchmark [26], which is on-par with 4-stage anchor-based Cascade R-CNN [2] and outperforms all existing anchor-free approaches. By adopting multi-scale training and testing (see Appendix for details), RPDet can achieve 46.5 AP, significantly surpassing all other detectors. Also note the AP_{50} of RPDet is relatively high, which is believed as a better metric in many applications [32].

6. Conclusion

In this paper, we propose *RepPoints*, a representation for object detection that models fine-grained localization information and identifies local areas significant for object classification. Based on *RepPoints*, we develop an object detector called RPDet that achieves competitive object detection performance without the need of anchors. Learning richer and more natural object representations like *RepPoints* is a direction that holds much promise for object detection.

	Backbone	Anchor-Free	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Faster R-CNN w. FPN [24]	ResNet-101		36.2	59.1	39.0	18.2	39.0	48.2
RefineDet [46]	ResNet-101		36.4	57.5	39.5	16.6	39.9	51.4
RetinaNet [25]	ResNet-101		39.1	59.1	42.3	21.8	42.7	50.2
Deep Regionlets [44]	ResNet-101		39.3	59.8	-	21.7	43.7	50.9
Mask R-CNN [13]	ResNeXt-101		39.8	62.3	43.4	22.1	43.2	51.2
FSAF [50]	ResNet-101		40.9	61.5	44.0	24.0	44.2	51.3
Cascade R-CNN [2]	ResNet-101		42.8	62.1	46.3	23.7	45.5	55.2
CornerNet [21]	Hourglass-104	✓	40.5	56.5	43.1	19.4	42.7	53.9
ExtremeNet [48]	Hourglass-104	✓	40.1	55.3	43.2	20.3	43.2	53.1
RPDet	ResNet-101	✓	41.0	62.9	44.3	23.6	44.1	51.7
RPDet	ResNet-101-DCN	✓	**42.8**	**65.0**	**46.3**	**24.9**	**46.2**	**54.7**
RPDet (ms train)	ResNet-101-DCN	✓	**45.0**	**66.1**	**49.0**	**26.6**	**48.6**	**57.5**
RPDet (ms train & ms test)	ResNet-101-DCN	✓	**46.5**	**67.4**	**50.9**	**30.3**	**49.7**	**57.1**

表7. 提出的RPDet与先前最先进检测器在COCO [26] test-dev数据集上的对比。提出的RPDet可实现46.5 AP，显著超越所有其他检测器。同时请注意RPDet的AP_{50}值相对较高，该指标在许多应用中被认为是更优的评估标准[32]。“ms”表示多尺度测试。

在边界框和RepPoints方面，基于中心点的方法分别超越了基于锚点的方法+1.1 mAP和+1.4 mAP，这很可能是因为它能更好地覆盖真实目标物体。

我们还将提出的基于RepPoints的无锚点检测器与RetinaNet[25]（一种流行的单阶段基于锚点的方法）、采用RoIAlign的FPN[24]（一种流行的两阶段基于锚点方法）[13]，以及从YOLOv1[31]的无锚点方法改进而来的类YOLO检测器进行比较，结果如表4所示。所提出的方法在性能上超越了RetinaNet[25]和FPN[24]方法，这两种方法均使用了每尺度多个锚点和复杂的锚点配置（FPN）。同时，所提出的方法也显著优于另一种无锚点方法（类YOLO检测器），分别以+4.4 mAP和+4.1 mAP的优势领先，这很可能归功于RepPoints表示的灵活性及其有效的优化机制。

将RepPoints转换为伪边界框。表5显示，第3.2节中提出的变换函数{v*}的不同实例化效果相当。

RepPoints与可变形RoI池化[4]起到互补作用。表6展示了将可变形RoI池化层[4]同时应用于边界框提议和RepPoints提议的效果。虽然对边界框提议应用可变形RoI池化层带来了+0.7 mAP的提升，但将其应用于RepPoints提议同样带来了+0.8 mAP的提升，这表明可变形RoI池化与所提出的RepPoints具有互补性。

5.3. RepPoints 可视化

在图4中，我们可视化了从COCO [26] minival数据集中选取的几个示例上学习到的*RepPoints*及其对应的检测结果。可以观察到，*Rep-Points*倾向于位于物体的极值点或关键语义点上。这些分布在物体上的点是自动学习得到的，无需显式监督。可视化结果也表明，本文提出的RPDet（此处采用最小-最大变换函数实现）能够有效检测微小物体。

5.4. 最先进技术对比

我们在测试开发集上将RPDet与最先进的检测器进行比较。表7展示了结果。在没有任何额外技巧的情况下，RPDet在COCO基准测试[26]上取得了42.8 AP，这与基于四阶段锚框的Cascade R-CNN[2]表现相当，并超越了所有现有的无锚框方法。通过采用多尺度训练和测试（详见附录），RPDet可以达到46.5 AP，显著超越所有其他检测器。同时值得注意的是，RPDet的{v*}相对较高，这被认为在许多应用中是更优的评估指标[32]。

6. 结论

本文提出了一种用于目标检测的表示方法*RepPoints*，该方法能够建模细粒度的定位信息，并识别对目标分类至关重要的局部区域。基于*RepPoints*，我们开发了一种名为RPDet的目标检测器，无需锚框即可实现具有竞争力的检测性能。学习如*RepPoints*般更丰富、更自然的目标表示，是目标检测领域极具前景的发展方向。

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Appendix

A1. Relationship between RepPoints and Deformable RoI pooling

In this section, we explain the differences between our method and deformable RoI pooling [4] in greater detail. We first describe the translation sensitivity of the regression step in the object detection pipeline. Then, we discuss how deformable RoI pooling [4] works and why it does not provide a geometric representation of objects, unlike the proposed RepPoints representation.

Translation Sensitivity We explain the translation sensitivity of the regression step in the context of bounding boxes. Denote a rectangular bounding box proposal before regression as $\mathcal{B}_P$ and the ground-truth bounding box as $\mathcal{B}_{GT}$. The target for bounding box regression can then be expressed as

$$T_P = \mathcal{F}(\mathcal{B}_P, \mathcal{B}_{GT}), \tag{7}$$

where $\mathcal{F}$ is a function for transforming $\mathcal{B}_P$ to $\mathcal{B}_{GT}$. This transformation is conventionally learned as a regression function $\mathcal{R}_B$:

$$\mathcal{R}_B(\mathcal{P}_B(I, \mathcal{B}_P)) = T_P = \mathcal{F}(\mathcal{B}_P, \mathcal{B}_{GT}), \tag{8}$$

where I is the input image and $\mathcal{P}_B$ is a pooling function defined over the rectangular proposal, e.g., direct cropping of the image [11], RoIPooling [33], or RoIAlign [13]. This formulation aims to predict the relative displacement to the ground truth box based on features within the area of $\mathcal{B}_P$. Shifts in $\mathcal{B}_P$ should change the target accordingly:

$$\mathcal{R}_B(\mathcal{P}_B(I, \mathcal{B}_P + \Delta\mathcal{B})) = \mathcal{F}(\mathcal{B}_P + \Delta\mathcal{B}, \mathcal{B}_{GT}). \tag{9}$$

Thus, the pooled feature $\mathcal{P}_B(I, B_P)$ should be sensitive to the box proposal B_P. Specifically, for any pair of proposals $\mathcal{B}_1 \neq \mathcal{B}_2$, we should have $\mathcal{P}_B(I, B_1) \neq \mathcal{P}_B(I, B_2)$. Most existing feature extractors $\mathcal{P}_B$ satisfy this property. Note that the improvement of RoIAlign [13] over RoIPooling [33] is partly due to this guaranteed translation sensitivity.

Analysis of Deformable RoI Pooling. For deformable RoI pooling [4], the system generates a pointwise deformation of samples on a regular grid [13] to produce a set of sample points S_P for each proposal. This can be formulated as

$$S_P = \mathcal{D}(I, \mathcal{B}_P), \tag{10}$$

where $\mathcal{D}$ is the function for generating the sample points. Then, bounding box regression aims to learn a regression function $\mathcal{R}_S$ which utilizes the sampled features via S_P to predict the target T_P as follows:

$$\mathcal{R}_S(\mathcal{P}_S(I, S_P)) = T_P = \mathcal{F}(\mathcal{B}_P, \mathcal{B}_{GT}) \tag{11}$$

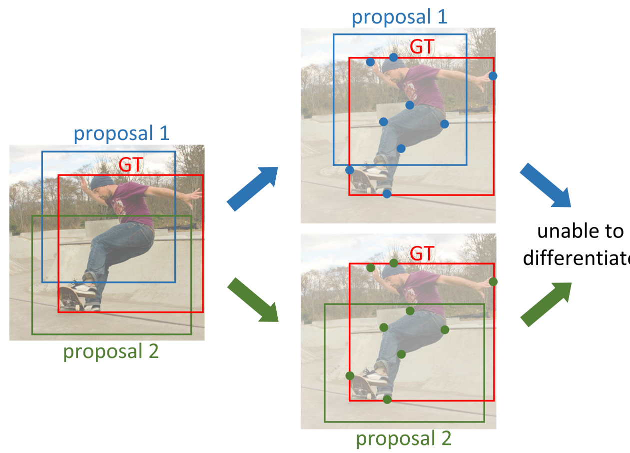

Figure 5. Illustration that deformable RoI pooling [4] is unable to serve as a geometric object representation, as discussed in Section 4 in the main paper. We consider two bounding box regressions based on different proposals. Assume that deformable RoI pooling [4] can learn a similar geometric object representation where the two sets of sample points lie at similar locations over the object of interest. For that to happen, the sampled features would need to be similar, such that the two proposals cannot be differentiated. However, deformable RoI pooling [4] can indeed differentiate nearby object proposals, leading to a contradiction. Thus, it is concluded that deformable RoI pooling [4] cannot learn the geometric representation of objects.

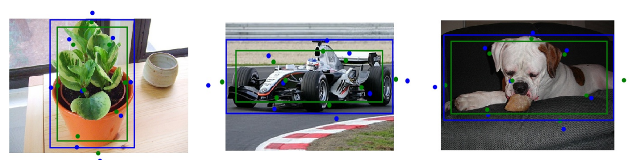

Figure 6. Visualization of the learned sample points of 3×3 deformable RoI pooling [4]. It is shown that the scale of sample points changes as the scale of the proposal changes, indicating that the sample points do not adapt to form a geometric *object* representation.

where $\mathcal{P}_S$ is the pooling function with respect to the sample points S_P.

From the translation sensitivity property, we have $\mathcal{P}_S(I, \mathcal{D}(I, \mathcal{B}_1)) \neq \mathcal{P}_S(I, \mathcal{D}(I, \mathcal{B}_2)), \forall \mathcal{B}_1 \neq \mathcal{B}_2$. Because the pooled feature $\mathcal{P}_S(I, \mathcal{D}(I, \mathcal{B}))$ is determined by the locations of sample points $\mathcal{D}(I, \mathcal{B})$, we have $\mathcal{D}(I, \mathcal{B}_1) \neq \mathcal{D}(I, \mathcal{B}_2), \forall \mathcal{B}_1 \neq \mathcal{B}_2$. This means that for two different proposals $\mathcal{B}_1$ and $\mathcal{B}_2$ of the same object, the sample points of these two proposals by deformable RoI pooling should be different. Hence, the sample points of different proposals cannot correspond to the geometry of the same object. They represent a property of the proposals rather than the geometry of the object.

Figure 5 illustrates the contradiction that arises if deformable RoI pooling were a representation of object ge-

附录

A1. RepPoints与可变形RoI池化之间的关系

在本节中，我们将更详细地阐述我们的方法与可变形RoI池化[4]之间的差异。首先，我们描述目标检测流程中回归步骤对平移的敏感性。接着，我们讨论可变形RoI池化[4]的工作原理，并解释为何它无法像我们提出的RepPoints表示法那样提供物体的几何表征。

翻译敏感性 我们在边界框的背景下解释回归步骤的翻译敏感性。将回归前的矩形边界框提议表示为 $\mathcal{B}_P$，将真实边界框表示为 $\mathcal{B}_{GT}$。那么，边界框回归的目标可以表示为

$$T_P = \mathcal{F}(\mathcal{B}_P, \mathcal{B}_{GT}), \tag{7}$$

其中 $\mathcal{F}$ 是将 $\mathcal{B}_P$ 转换为 $\mathcal{B}_{GT}$ 的函数。这种转换通常被学习为一个回归函数 $\mathcal{R}_B$:

$$\mathcal{R}_B(\mathcal{P}_B(I, \mathcal{B}_P)) = T_P = \mathcal{F}(\mathcal{B}_P, \mathcal{B}_{GT}), \tag{8}$$

其中 I 是输入图像，$\mathcal{P}_B$ 是在矩形建议区域上定义的池化函数，例如直接裁剪图像 [11]、RoIPooling [33] 或 RoIAlign [13]。该公式旨在基于 $\mathcal{B}_P$ 区域内的特征预测相对于真实边界框的位移。$\mathcal{B}_P$ 的变化应相应地改变目标：

$$\mathcal{R}_B(\mathcal{P}_B(I, \mathcal{B}_P + \Delta\mathcal{B})) = \mathcal{F}(\mathcal{B}_P + \Delta\mathcal{B}, \mathcal{B}_{GT}). \tag{9}$$

因此，池化特征 $\mathcal{P}_B(I, B_P)$ 应对候选框 B_P 保持敏感。具体而言，对于任意一对候选框 $\mathcal{B}_1 \neq \mathcal{B}_2$，我们应满足 $\mathcal{P}_B(I, B_1) \neq \mathcal{P}_B(I, B_2)$。大多数现有特征提取器 $\mathcal{P}_B$ 均符合这一特性。值得注意的是，RoIAlign [13] 相较于 RoIPooling [33] 的改进，部分原因正在于这种有保证的平移敏感性。

可变形RoI池化分析。对于可变形RoI池化[4]，系统通过在规则网格[13]上生成逐点形变，为每个候选区域生成一组采样点S_P。其公式可表述为

$$S_P = \mathcal{D}(I, \mathcal{B}_P), \tag{10}$$

其中$\mathcal{D}$是生成采样点的函数。随后，边界框回归旨在学习一个回归函数$\mathcal{R}_S$，该函数通过S_P利用采样特征来预测目标T_P，具体如下：

$$\mathcal{R}_S(\mathcal{P}_S(I, S_P)) = T_P = \mathcal{F}(\mathcal{B}_P, \mathcal{B}_{GT}) \tag{11}$$

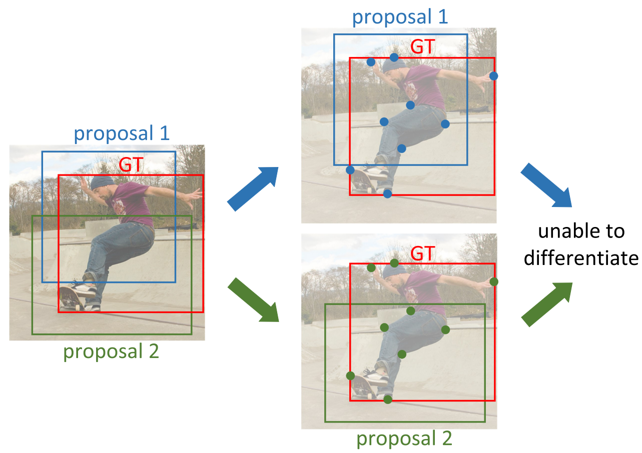

图5. 如主论文第4节所述，可变形RoI池化[4]无法作为几何物体表征的示意图。我们考虑基于不同候选框的两种边界框回归。假设可变形RoI池化[4]能够学习到相似的几何物体表征，使得两组采样点位于目标物体的相似位置。要实现这一点，采样特征必须相似，从而导致两个候选框无法被区分。然而，可变形RoI池化[4]实际上能够区分相邻的物体候选框，由此产生矛盾。因此可得出结论：可变形RoI池化[4]无法学习物体的几何表征。

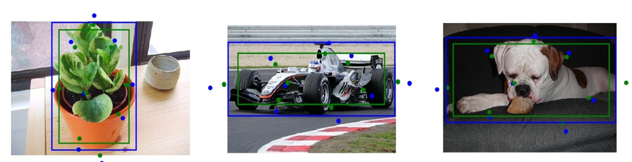

图6. 3×3可变形RoI池化[4]学习到的采样点可视化。结果表明，采样点的尺度随提案框的尺度变化而改变，这表明采样点未能自适应地形成几何*object*表示。

其中$\mathcal{P}_S$是关于样本点S_P的池化函数。

根据翻译敏感性特性，我们有$\mathcal{P}_S(I, \mathcal{D}(I, \mathcal{B}_1)) \neq \mathcal{P}_S(I, \mathcal{D}(I, \mathcal{B}_2)), \forall \mathcal{B}_1 \neq \mathcal{B}_2$。由于池化特征$\mathcal{P}_S(I, \mathcal{D}(I, \mathcal{B}))$由采样点$\mathcal{D}(I, \mathcal{B})$的位置决定，因此我们得到$\mathcal{D}(I, \mathcal{B}_1) \neq \mathcal{D}(I, \mathcal{B}_2), \forall \mathcal{B}_1 \neq \mathcal{B}_2$。这意味着对于同一物体的两个不同提议$\mathcal{B}_1$和$\mathcal{B}_2$，可变形RoI池化为这两个提议生成的采样点应当不同。因此，不同提议的采样点无法对应同一物体的几何结构。它们代表的是提议本身的属性，而非物体的几何特征。

图5展示了如果可变形RoI池化是物体几何形状的表示，将会产生的矛盾。

method	backbone	ms train	ms test	AP
RPDet	R-50			38.6
	R-50	✓		40.8
	R-50	✓	✓	42.2
	R-101			40.3
	R-101	✓		42.3
	R-101	✓	✓	44.1
	R-101-DCN			43.0
	R-101-DCN	✓		44.8
	R-101-DCN	✓	✓	46.4
	X-101-DCN			44.5
	X-101-DCN	✓		45.6
	X-101-DCN	✓	✓	46.8

Table 8. Benchmark results of RPDet on MS-COCO [26] validation set (`minival`). All the models here are trained with FPN [24] under the '2x' setting [12]. For the backbone notation, 'R-50' and 'R-101' denotes ResNet-50 and ResNet-101 [15] respectively. 'R-101-DCN' denotes ResNet-101 with all convolution layers substituted with deformable convolution layers [4]. 'X' denotes the ResNeXt-101 [43] backbone. "ms" indicates multi-scale.

ometry. Moreover, Figure 6 illustrates that, for the learned sample points of two proposals for the same object by deformable RoI pooling, the sample points represent a property of the proposals instead of the geometry of the object.

RepPoints In contrast to deformable RoI pooling where the pooled features represent the original bounding box proposals, the features extracted from RepPoints localize the object. As it is not restricted by translation sensitivity requirements, RepPoints can learn a geometric representation of *objects* when localization supervision on the corresponding pseudo box is provided (see Figure 4 in the main paper). While object localization supervision is not applied on the sample points of deformable RoI pooling, we show in Table 2 in the main paper that such supervision is crucial for RepPoints.

It is worth noting that deformable RoI pooling [4] is shown to be complementary to the RepPoints representation (see Table 6 in the main paper), further indicating their different functionality.

A2. More Benchmark Results for RPDet

We present more benchmark results of our proposed detector RPDet in Table 8. Our PyTorch implementation is available at https://github.com/microsoft/RepPoints. All models were tested on MS-COCO [26] validation set (`minival`).

Multi-scale training and test settings. In multi-scale training, for each mini-batch, the shorter side is randomly selected from a range of $[480, 960]$. In multi-scale testing, we first resize each image to a shorter side of $\{400, 600, 800, 1000, 1200, 1400\}$. Then the detection results (before NMS) from all scales are merged, followed by a NMS step to produce the final detection results.

method	backbone	ms train	ms test	AP
RPDet	R-50			38.6
	R-50	✓		40.8
	R-50	✓	✓	42.2
	R-101			40.3
	R-101	✓		42.3
	R-101	✓	✓	44.1
	R-101-DCN			43.0
	R-101-DCN	✓		44.8
	R-101-DCN	✓	✓	46.4
	X-101-DCN			44.5
	X-101-DCN	✓		45.6
	X-101-DCN	✓	✓	46.8

表8. RPDet在MS-COCO [26]验证集（minival）上的基准测试结果。此处所有模型均采用FPN [24]在‘2x’设置[12]下训练。主干网络标注中，‘R-50’和‘R-101’分别表示ResNet-50和ResNet-101[15]。‘R-101-DCN’表示将所有卷积层替换为可变形卷积层[4]的ResNet-101。‘X’表示ResNeXt-101[43]主干网络。“ms”代表多尺度测试。

此外，图6表明，通过可变形RoI池化学习到的同一物体两个提议的采样点，这些采样点代表的是提议的属性，而非物体的几何结构。

RepPoints 与可变形RoI池化不同，后者池化后的特征代表原始边界框提议，而RepPoints提取的特征直接定位物体。由于不受平移敏感性要求的限制，当提供对应伪框的定位监督时（参见主论文图4），RepPoints能够学习到{v*}的几何表示。虽然物体定位监督未应用于可变形RoI池化的采样点，但我们在主论文表2中证明，这种监督对RepPoints至关重要。

值得注意的是，可变形RoI池化[4]被证明与RepPoints表示具有互补性（参见主论文表6），进一步表明它们具有不同的功能。

A2. RPDet的更多基准测试结果

我们在表8中展示了我们提出的检测器RPDet的更多基准测试结果。我们的PyTorch实现可在https://github.com/microsoft/RepPoints获取。所有模型均在MS-COCO [26]验证集（minival）上进行了测试。

Multi-scale training and test settings. 在多尺度训练中，每个小批次的图像短边会从[480, 960]范围内随机选取。在多尺度测试中——

首先，我们将每张图像的短边调整为{400, 600, 800, 1000, 1200, 1400}。接着，合并所有尺度下的检测结果（在非极大值抑制之前），最后通过一个非极大值抑制步骤生成最终的检测结果。